Lab 12:

Character Embeddings & Dimensionality Reduction

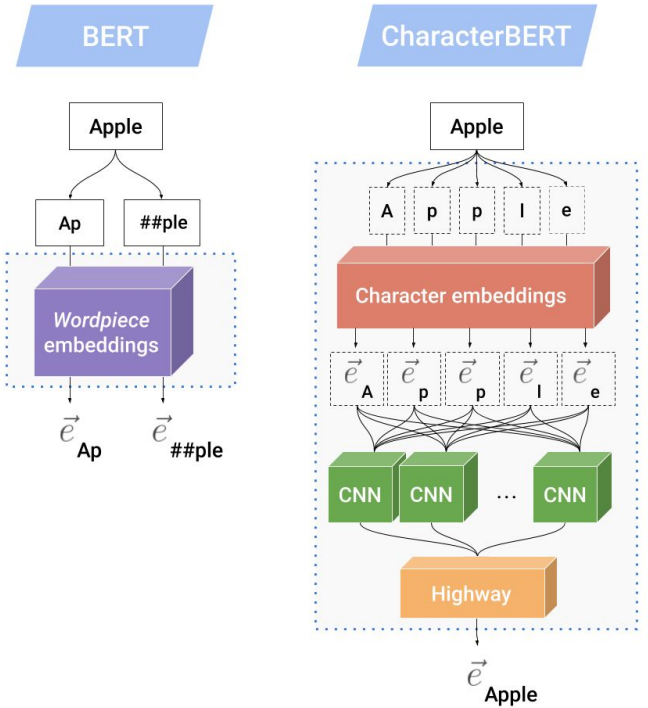
Today’s Agenda

1. Review
    - Character Embeddings
    - Dimensionality Reduction
        - PCA
        - Autoencoders

2. Jupyter Notebook Examples

Character Embeddings

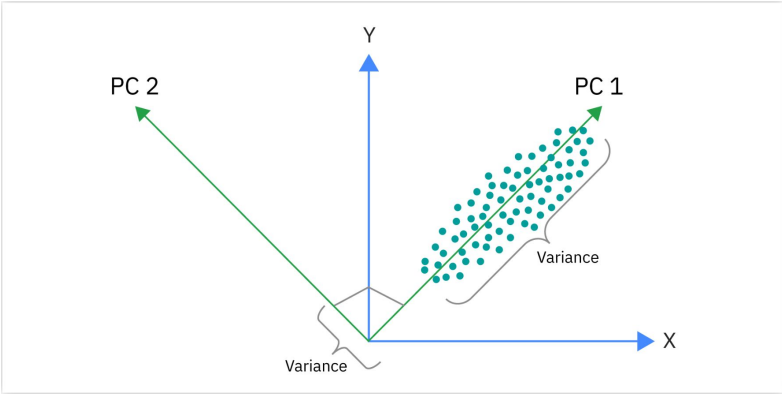
- Each character in a word represented as a individual dense vector
- Words = sequences of character vectors
- Sequence of vectors can be processed with RNNs/LSTMs, CNNs, Transformers
- Reduces vocabulary size
- Great at learning subword patterns, helpful for rare or mispelled words



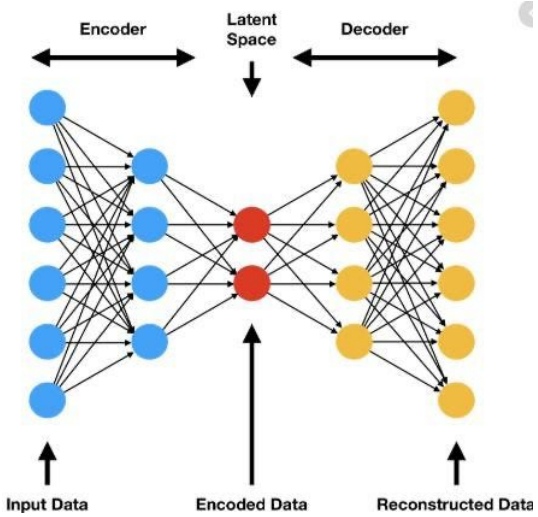
Source: https://towardsdatascience.com/characterbert-reconciling-elmo-and-bert-for-word-level-open-vocabulary-representations-from-94037fe68b21/

Dimensionality Reduction

- Reduce number of features, while retaining important details
- Two procedures:
  - Feature selection: Selects the most important features
  - Feature projection: Creates new features by transforming the original ones into a lower-dimensional space



Source: https://www.ibm.com/think/topics/principal-component-analysis



Source: https://medium.datadriveninvestor.com/dimensionality-reduction-using-an-autoencoder-in-python-bf540bb3f085

Comparison

Principal Component Analysis (PCA)

- Project data into a lower-dimensional space using **principal components** (directions of max variance)
- Use **eigenvectors** (principal components) and **eigenvalues** (variance amount) from covariance matrix
- Goal: Maximize variance using orthogonal directions
- Better for linear relationships

Autoencoders

- Type of NN with:
  - **Encoder** to compress input data (image) to lower dimension
  - **Decoder** to reconstruct the original input from reduced encoding
- Goal: Minimize MSE between the reconstructed image and the original image
- Better for nonlinear relationships

Example Notebook